# 14 Year-end review and consolidation of Seconde learning

**Sheet 80** Mon 14 June 2027 *Year-end review and consolidation*

1. Warm-up · Quick [GEO] $A(-1;4)$ and $B(3;1)$. Coordinates of $\overrightarrow{AB}$?

   **Ans.** $\begin{pmatrix} 4 \\ -3 \end{pmatrix}$.

2. Core · Method [GEO] Equation of $(AB)$ in gradient-intercept form with $A(0;1)$ and $B(2;5)$.

   **Ans.** Gradient 2; $y = 2x + 1$.

3. Core + · Method [REP] $A(1;0)$, $B(4;1)$, $C(5;4)$. Find $D$ so that $ABCD$ is a parallelogram.

   **Ans.** $D = A + C - B = (2;3)$; check: $\overrightarrow{AB} = \overrightarrow{DC} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$.

4. Review · Quick [CAL] Calculate $-5 + 8 \times (-2)$.

   **Ans.** $-5 - 16 = -21$.

   *Multiplication takes priority.*

5. Thread · Statement → formula [REG] The area of a square of side $x$ is a function $A$ of $x$. Give $A(x)$, a table of values for $x = 1\,;\,2\,;\,3$, and say whether $A$ is proportional to $x$.

   **Ans.** $A(x) = x^2$; $1\,;\,4\,;\,9$; not proportional: $A(2) \neq 2A(1)$.

**Sheet 81** Tue 15 June 2027 *Year-end review and consolidation*

1. Warm-up · Quick [NUM] Write 0.00045 in standard form.

   **Ans.** $4.5 \times 10^{-4}$.

2. Core · Quick [NUM] Find $[-2;3[ \cap ]0;5]$.

   **Ans.** $]0;3[$.

3. Core + · Method [EQ] Solve $|x - 1| < 4$ and give $S$ as an interval.

   **Ans.** $-4 < x - 1 < 4$: $S = ]-3;5[$.

4. Review · Expand [DEV] Expand and simplify $(2x - 1)^2$.

   **Ans.** $4x^2 - 4x + 1$.

5. Thread · Quick [GEO] A cylinder has radius 3 cm and height 5 cm. Calculate its exact volume.

   **Ans.** $V = \pi \times 3^2 \times 5 = 45\pi$ cm$^3$.

**Sheet 82** Thu 17 June 2027 *Year-end review and consolidation*

1. Warm-up · Quick [IMG] $f(x) = x^2 - 4x$. Calculate $f(3)$.

   **Ans.** $9 - 12 = -3$.

2. Core · Quick [EQ] Solve $x^2 = 9$ and then $|x| = 2$.

   **Ans.** $x = \pm 3$; $x = \pm 2$.

3. Core + · Method [REG] $f(x) = 2x - 3$ and $g(x) = x + 1$. Solve $f(x) < g(x)$.

   **Ans.** $2x - 3 < x + 1 \Leftrightarrow x < 4$: $S = ]-\infty;4[$.

4. Review · Quick [QP] A tap delivers 18 L in 4 min. What volume does it deliver in 10 min?

   **Ans.** $4.5 \times 10 = 45$ L.

5. Thread · Reading code [ALG] What is the final value of `n`?

```
n = 0
while n*n < 30:
    n = n + 1
```

   **Ans.** $n = 6$ because $5^2 = 25 < 30$ but $6^2 = 36 \geqslant 30$.

   *Smallest integer whose square exceeds* 30.

**Sheet 83** Mon 21 June 2027 *Year-end review and consolidation*

1. Warm-up · Quick [STA] Mean of 5, 7, 9, 11?
   **Ans.** 8.
2. Core · Method [STA] Data set 1 ; 3 ; 4 ; 6 ; 8 ; 9 ; 11: find $Q_1$, the median and $Q_3$.
   **Ans.** $n = 7$: $Q_1$ has rank $\lceil 7/4 \rceil = 2$: 3; median (rank 4): 6; $Q_3$ has rank $\lceil 21/4 \rceil = 6$: 9.
3. Core + · True/false [STA] True or false? Justify. Adding 10 to every value of a data set leaves its standard deviation unchanged.
   **Ans.** True. The deviations from the mean do not change.
4. Review · Reflection [IMG] How many preimages do $-1$ and 4 have for $f(x) = x^2$?
   **Ans.** $-1$: none; 4: two ($-2$ and 2).
5. Thread · Prove [LIT] Show that for every real $x$: $(x-1)(x+1)+1 = x^2$.
   **Ans.** $(x-1)(x+1)+1 = x^2 - 1 + 1 = x^2$.

**Sheet 84** Tue 22 June 2027 *Year-end review and consolidation*

1. Warm-up · Quick [DEV] Expand and simplify $(2x-3)(x+4)$.
   **Ans.** $2x^2 + 8x - 3x - 12 = 2x^2 + 5x - 12$.
2. Core · Quick [EQ] Solve $3x - 7 \leqslant 2x + 5$.
   **Ans.** $x \leqslant 12$; $S = ]-\infty; 12]$.
3. Core + · Method [EQ] Solve $(x-1)(x+4) = 0$ and then $x^2 = 5$.
   **Ans.** $x = 1$ or $x = -4$; $x = \sqrt{5}$ or $x = -\sqrt{5}$.
4. Review · Reflection [PNP] Membership: 5 € + 3 € per entry. Is the price proportional to the number of entries?
   **Ans.** No: 1 entry $\mapsto$ 8 €, 2 entries $\mapsto$ 11 $\neq$ 16 €.
5. Thread · Quick [SOL] A cone has radius 3 cm and height 4 cm. Calculate its slant height and its exact volume.
   **Ans.** Slant height $\sqrt{3^2 + 4^2} = 5$ cm; $V = \dfrac{1}{3}\pi \times 9 \times 4 = 12\pi$ cm$^3$.

**Sheet 85** Thu 24 June 2027 *Year-end review and consolidation*

1. Warm-up · Quick [PRO] Which multiplier corresponds to an increase of 8 %?
   **Ans.** 1.08.
2. Core · Method [PRO] An increase of 25 % is followed by a decrease of 20 %. What is the overall rate of change?
   **Ans.** $1.25 \times 0.8 = 1$: no change (0 %).
3. Core + · Quick [PRO] A price goes from 5 000 to 5 750 Ar. Rate of change?
   **Ans.** $\dfrac{750}{5\,000} = 15\,\%$.
4. Review · Recognise [PER] What is the cross-section of a cylinder by a plane containing its axis? By a plane perpendicular to its axis?
   **Ans.** A rectangle; a disc.
5. Thread · Problem [EQ] Three times a number, minus 4, is equal to twice that number plus 5. Write an equation and then find the number.
   **Ans.** $3x - 4 = 2x + 5$, so $x = 9$.
   *Check:* $3 \times 9 - 4 = 23 = 2 \times 9 + 5$.

**Sheet 86** Mon 28 June 2027 *Year-end review and consolidation*

1. Warm-up · Quick [MOD] Direction of variation of $x \mapsto -3x+2$?
   **Ans.** Strictly decreasing (gradient $-3<0$).
2. Core · Method [REG] Minimum and maximum of $f(x)=x^2$ on $[-3;1]$?
   **Ans.** Minimum 0 (at 0); maximum 9 (at $-3$); $f(1)=1<9$.
3. Core + · Method [CMP] For $x>0$, compare $\dfrac{1}{x}$ and $\dfrac{1}{x+1}$.
   **Ans.** $0<x<x+1$ and the reciprocal function is decreasing on $]0;+\infty[$: $\dfrac{1}{x}>\dfrac{1}{x+1}$.
4. Review · Construction [CON] Describe in three steps the construction of the reflection $A'$ of $A$ in a line $(d)$.
   **Ans.** 1. Draw the perpendicular to $(d)$ through $A$, meeting $(d)$ at $H$. 2. Using compasses, mark off $HA$ on the other side of $(d)$. 3. $A'$ is the point obtained: $H$ is the midpoint of $[AA']$.
5. Thread · Method [DEM] A point $I$ satisfies $IA=IB$. Where does it lie? Which property is used?
   **Ans.** On the perpendicular bisector of $[AB]$ (characterisation of the perpendicular bisector).

**Sheet 87** Tue 29 June 2027 *Year-end review and consolidation*

1. Warm-up · Quick [PRB] $P(A)=0.28$. Calculate $P(\overline{A})$.
   **Ans.** 0.72.
2. Core · Method [PRB] 100 people, including 60 women; 6 women and 4 men are left-handed. Calculate $P_F(G)$ and $P_G(F)$.
   **Ans.** $P_F(G)=\dfrac{6}{60}=0.1$; $P_G(F)=\dfrac{6}{10}=0.6$.
3. Core + · Method [PRB] $P(A)=0.4$, $P_A(B)=0.5$, $P_{\overline{A}}(B)=0.2$. Calculate $P(B)$.
   **Ans.** $0.4\times0.5+0.6\times0.2=0.2+0.12=0.32$.
4. Review · Problem [MOD] Volume of a cube of edge $x$: $V(x)=x^3$. Calculate $V(3)$ and find $x$ such that $V(x)=64$.
   **Ans.** $V(3)=27$; $x^3=64$, $x=4$.
5. Thread · Quick [REP] In which quadrant (numbered I to IV anticlockwise) does $M(a;b)$ lie, with $a<0$ and $b>0$?
   **Ans.** Quadrant II (top left).

**Sheet 88** Thu 1st July 2027 *Year-end review and consolidation*

1. Warm-up · Quick [GEO] Hypotenuse of a right-angled triangle with sides 5 and 12?
   **Ans.** $\sqrt{25+144}=13$.
2. Core · Quick [REP] Midpoint of $[AB]$ with $A(-3;2)$ and $B(5;-4)$?
   **Ans.** $\left(\dfrac{-3+5}{2};\dfrac{2-4}{2}\right)=(1;-1)$.
3. Core + · Write-up [DEM] Show that $A(0;0)$, $B(2;1)$, $C(-1;2)$ form an isosceles right-angled triangle at $A$.
   **Ans.** $AB^2=5$, $AC^2=5$; $BC^2=9+1=10=AB^2+AC^2$: converse of Pythagoras' theorem.
4. Review · Quick [NUM] Give an approximate value of $\dfrac{1}{3}$ correct to $10^{-2}$. Is this number a decimal?
   **Ans.** 0.33. No: the decimal expansion of $\frac{1}{3}$ is unending ($0.333\ldots$).
5. Thread · Problem [DIV] What is the smallest integer that leaves remainder 1 when divided by 4 and by 6, other than 1?

**Ans.** $1 + \text{PPCM}(4\,;\,6) = 13$.